

AI Assignment: Spartans Chess Tournament

Assignment 1

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Instructions for Submission

- **Deadline:** Sept. 13, 2026 11:59 PM; No extension will be given.
- **Code Submission:** You must submit a single Python file named [YourRollNumber].py (in all uppercase). For example, *P22CS201.py*.

Important Notes:

- **File Type:** Only a single Python (.py)
- **Code File:** All of your agent's code must be contained within a single Python file. You must implement your solution by modifying the provided template file, `ai_player.py`. Before submission, rename the file to [YourRollNumber].py (**all uppercase**), and update **the class name inside the file to [YourRollNumber] (all uppercase)**.

Important: For example: if your roll no. is P22CS201, the file you submit should be named P22CS201.py, and the class defined inside should be named P22CS201.

The template for the AI agent is provided in the "ai_player.py" file on GitHub.

- **Evaluation:**
 - Code: 10, Ranking: 10, Viva:20.
 - If your agent is not able to beat the sample players provided, the ranking component will get a score of 0.
 - Submissions that do not adhere to these exact naming conventions and file requirements will not be evaluated and will automatically receive a score of zero & no reconsideration will be entertained.
- This is a competition and an open-ended problem. The better your solution, the better your performance will be.
- Zero marks will be awarded for the **Code** and **Ranking** components of the assignment in the following cases:
 - Submission of a corrupted Python file or an agent's code that is non-functional.
 - Plagiarism in any form among students.
 - Any form of tampering will result in an immediate score of 0 in the entire assignment.
- **Failure to appear for the viva will result in a score of 0 for the assignment in its entirety.**

1 Spartans 300: Code only

You can access the assignment template here: [Assignment 1 GitHub Repository](#) . Please refer to the repository **README** for all instructions, documents, guidelines, and answers to common doubts. In case you have any further questions or encounter an issue that is not addressed in the README, please **create an issue in the repository** following the procedure described there.

This is a chess tournament where all participants' agents will compete against each other. After the submission deadline, we will run a round-robin tournament, and each pair of agents will play two matches one as white and one as black.

1.1 Objective

The primary objective of this assignment is to develop a mini 6x8 chess game. Utilizing principles of Adversarial Search, you will build a robust agent capable of competing effectively against other agents. Your solution should aim to win by intelligently evaluating board states and making the optimal moves to outperform opponents. This task emphasizes practical application of adversarial search algorithms and evaluation functions in a competitive, open-ended problem.

Note: We highly encourage you to go through the entire code base so that you can have a better understanding and can create a better AI agent.

Note: Do not modify the code for the underlying chess engine, since the evaluation will be done with the original copy of the chess engine. Your only contribution will be the AI agent.

Note: The agent that you provide has to be an adversarial agent that decides the next move based on a game tree. You can make it faster using various techniques including those discussed in class. **DONOT** use reinforcement learning-based agents for this problem.

1.2 Game Rules and Constraints

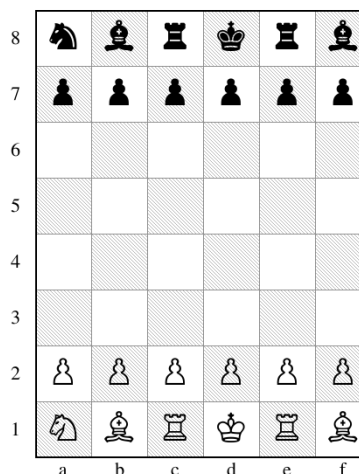


Figure 1: 6x8 Chess Board

Chess Rules

1. Total of 12 pieces per side: 1 knights, 2 bishop, 2 rooks, 1 king, and 4 pawns (see Fig. 1).
2. All pieces follow standard chess movement rules, but pawns are limited to a single-step move; all other special moves are illegal.
3. Game Type: (60-second game). The game will have a turn limit of 150 for the entire game (75 for each player).
4. In this game, no special rules or moves are allowed. Examples include the following. No such moves will be allowed.
 - En passant
 - Castling
 - Pawn promotion

Constraints

1. You must either draw or win the game:
 - (a) Time cap: 60 seconds.
 - (b) Maximum turns per agent: 75 moves
2. If any player exceeds the 60-second time limit, they will automatically lose the match and be awarded a score of 0 for that game.

Points

Note: We will evaluate based on checks and capturing pieces.

1. For each check= +2
2. Capturing pawn = +20
3. Capturing Bishop = +70
4. Capturing Knight = +70
5. Capturing Rook = +100
6. Checkmate (Capturing King) = +600

Match Score Card

Note: The final score of the game will be determined as follows:

1. **Checkmate:** Achieving checkmate awards an absolute score of **+600**, overriding any points from captured pieces.
2. **Loss:** A loss results in a score of **0**, regardless of prior captures.
3. **Draw/Stalemate:** In a draw/stalemate, the final score is calculated exclusively from the points earned by capturing pieces as per the Points table.

Word of Advice

Note: In last year's iteration, **25 students were caught tampering with the game**, and **15 cases of plagiarism were detected**. This year, the evaluation process will be significantly stricter, with a strong emphasis on maintaining the integrity and fairness of the assignment. A number of students previously attempted to identify loopholes in the system and gain an unfair advantage, but failed miserably. Such attempts were detected and dealt with accordingly. We strongly advise you **not to attempt any form of tampering, exploitation, plagiarism, or manipulation of the game or evaluation system**.

Any such instance detected this year will result in an **immediate score of 0 in the assignment component**, regardless of the student's performance or score in any other part of the assignment.

Please play fair, follow the rules, and focus on solving the assignment as intended.